Alkanes, Alkenes, and Alkynes

Ingyu Bahng

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Contents

1 Alkenes

Core Concept 1.1: Alkene Orbital Structure

The orbital structure of an alkene double bond arises from the combination of a σ bond and a π bond. This additional π overlap results in a shorter bond length relative to an alkane single bond: whereas a typical C–C single bond measures roughly 1.5 A, the C=C double bond is contracted to approximately 1.3 A.

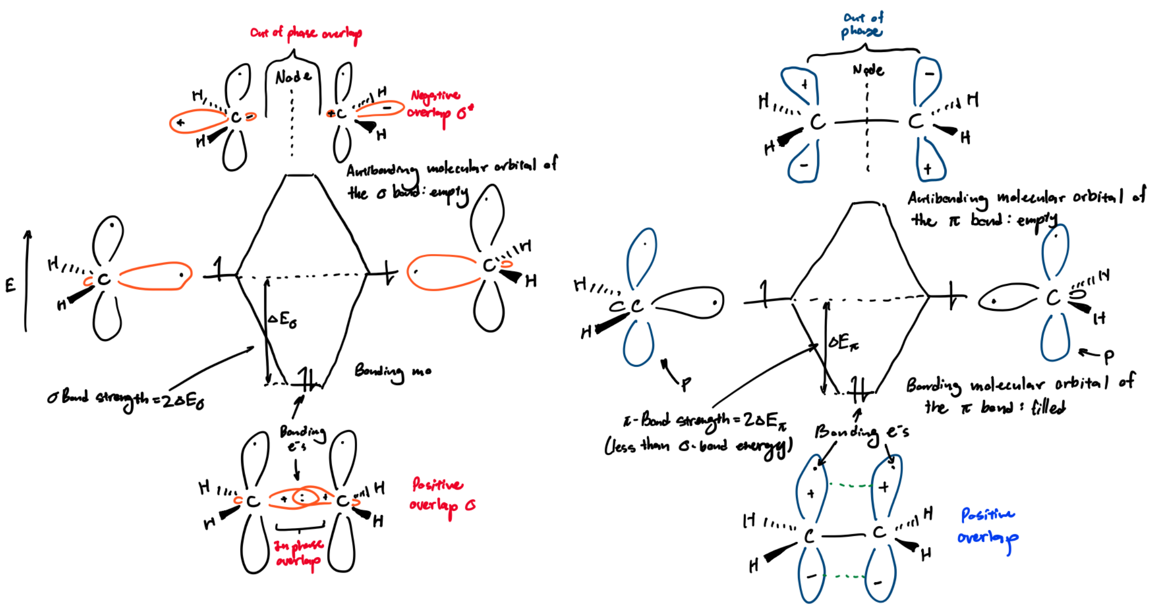


Figure 1: Alkene σ (*left*) and π (*right*) bonding orbitals

Core Concept 1.2: Alkene Bond Strength

The π bond of an alkene is relatively weaker than its partner σ bond because the antibonding π^* orbital sits closer in energy to the bonding π orbital than the antibonding σ^* orbital sits to the bonding σ orbital. As a result, breaking an alkene π bond requires roughly 65 $kcal\ mol^{-1}$, whereas breaking the σ bond requires 108 $kcal\ mol^{-1}$.

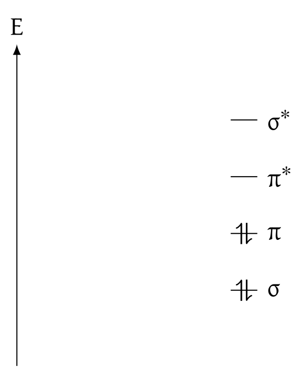
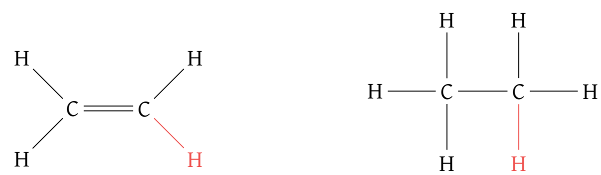


Figure 2: Molecular bonding orbitals levels of σ, π, π^*, and σ^* in an alkene double bond; *notice that the distance between π and π^* is less than that of σ and σ^**

By comparison, the typical alkane σ C–C bond strength is roughly 88 $kcal\ mol^{-1}$. The alkene σ bond is therefore stronger than the alkane σ bond, since the alkene σ bond is formed from sp^2 hybridized orbitals rather than the sp^3 orbitals used in alkanes. This **greater s-orbital character**—33% versus 25%, respectively—corresponds to greater bond strength, as the bonding e^-s are held closer to the nucleus relative to p orbitals.

Consequently, when comparing the pK_a values of the substituent hydrogens on alkenes and alkanes, alkenes exhibit lower pK_a values than alkanes.



$pK_a \approx 44$ $pK_a \approx 50$

Figure 3: Ethene (*left*) vs ethane (*right*) pK_a values

In principle, then, the reaction between methyllithium and an alkene below will proceed to a product-favoring equilibrium, since the alkene is more acidic (*better able to release a proton*) than the alkane.

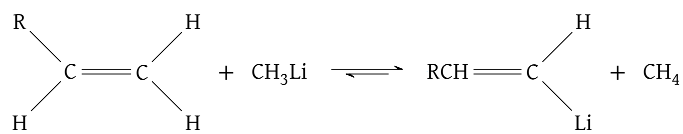


Figure 4: Methyllithium reaction showcasing alkene favoring equilibrium due to pK_a differences

In practice, however, the reaction shown in Figure 4 faces regio- and stereoselectivity issues, since the remaining substituent hydrogens may dissociate equally or more favorably than the one depicted. To overcome these problems, chemists instead employ the formation of **organometallics** (*Core Concept 2.3*).

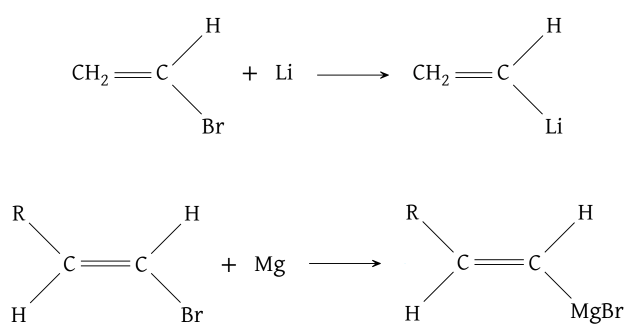


Figure 5: Methyllithium reaction showcasing alkene favoring equilibrium due to pK_a differences

Core Concept 1.3: Relative Stability of Alkenes

Consider the **heat of hydrogenation ($\Delta H_{hydrogenation}$)** of the following butene isomers upon conversion to butane.

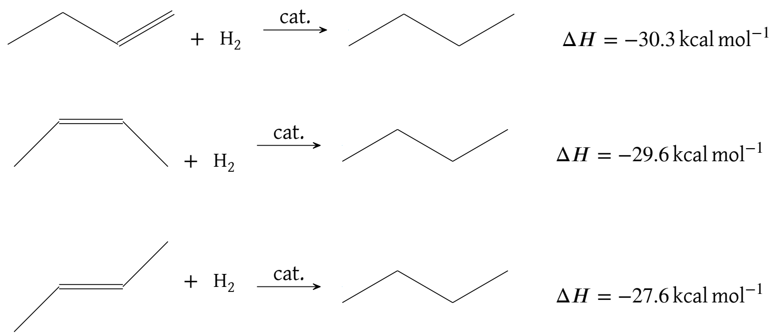


Figure 6: Butane isomer conversions to butane

The greater relative stability of internal double bonds over terminal ones, and of trans over cis substituents, arises from two factors.

(a) **Hyperconjugation** allows the temporary charge imbalance arising in the transition state to be better distributed among neighboring orbitals in the internal bond butene than in its terminal counterpart.

(b) **Steric hindrance** between the cis methyl groups increases the relative instability of the cis butene compared to its trans counterpart.

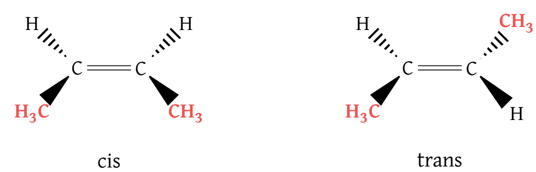


Figure 7: Steric hinderance between methyl groups of cis (*left*) vs trans (*right*) butene; *more steric exists between methyl groups of cis butene because of closer methyl group proximity*

The general order of alkene stability thus follows as shown below.

$$CH_2{=}CH_2 \quad < \quad RCH{=}CH_2 \quad < \quad RCH{=}CHR_{cis} \quad < \quad RCH{=}CHR_{trans} \quad < \quad RCH{=}CR_2 \quad < \quad R_2C{=}CR_2$$

1.1 Alkene Synthesis

Core Concept 1.4: Elimination & Saytzev Rule

Synthesizing alkenes from alkanes through E2 elimination requires the consideration of several factors.

(a) **Base Size**

Consider an E2 reaction **with a small base** on the following RX alkane.

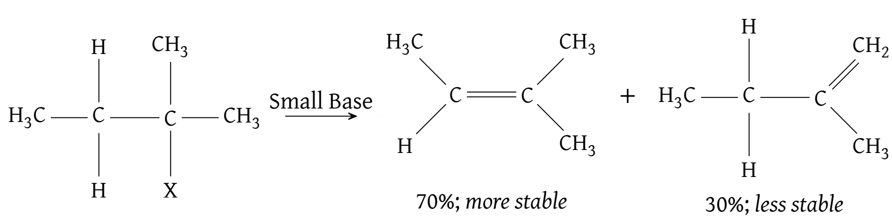


Figure 8: E2 reaction on a haloalkane using a small base

Notice that the internal double bond alkene is favored over the terminal version, since the advantages of hyperconjugation outweigh the minimal steric hindrance associated with a small base. In contrast, an E2 reaction **using a large base** (*Figure 9*) favors the terminal version, since steric hindrance exerts a comparatively larger influence on transition state stability, sufficient to override the hyperconjugative favorability of the internal double bond product.

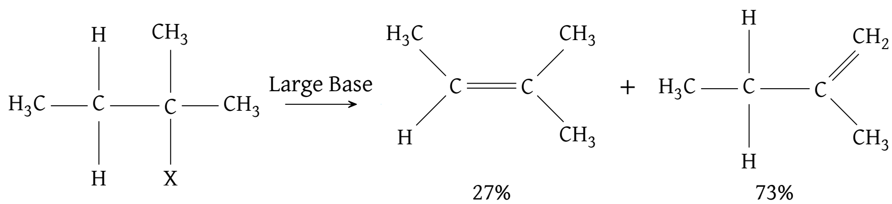


Figure 9: E2 reaction on a haloalkane using a large base

This phenomenon is termed the **Saytzev Rule**, which states that non-bulky bases lead to the more stable internal double bond product. The corresponding energy diagrams of the two reactions are shown below to depict the energy levels of the reaction participants throughout the reaction.

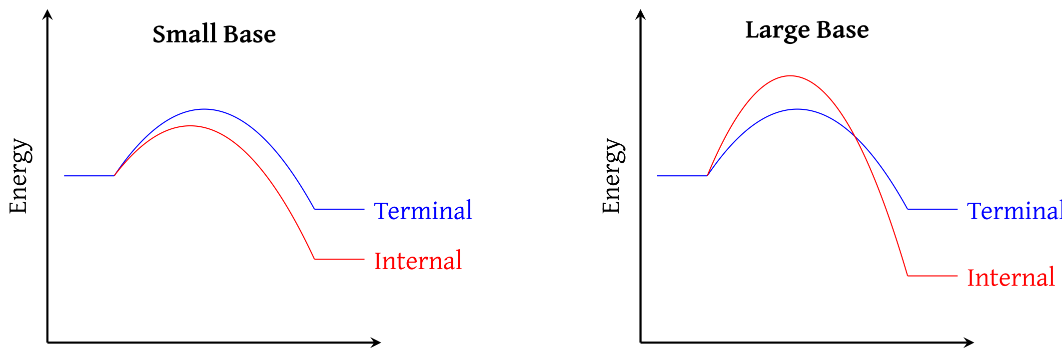


Figure 10: Energy diagrams of haloalkane elimination reactions producing alkenes; [Small Base] *lower energy transition state (red) favors formation of more stable product*; [Large Base] *with a large base, the lower energy transition state (blue) now favors foramation of less stable product*

(b) **Stereoselectivity**

The presence of **multiple stereoisomers**—particularly among the internal double bond alkenes formed via E2 reactions with small bases—is another factor a chemist must consider. Although the E2 reaction is stereoselective, favoring the more stable trans isomers over the sterically hindered cis isomers, this preference is not absolute (*some cis isomers do form*).

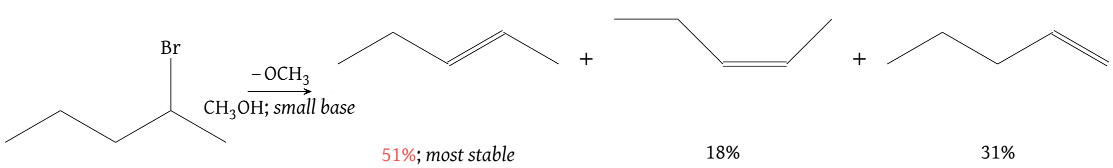


Figure 11: Alkene product mixture from E2 elimination of 2-bromo-pentane

(c) **Other Factors**

Other factors, such as preferable precursor conformations—namely rotamers—play a role as well.

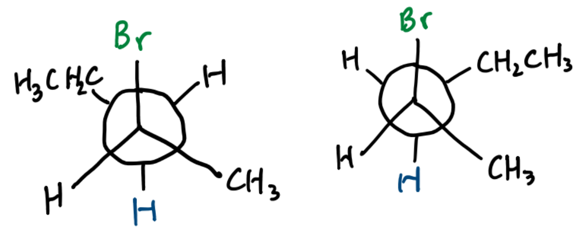


Figure 12: Precursor conformations of 2-bromo-pentane; *the left rotamer conformation is more favorable because the ethyl and methyl substitutents are anti to one another whereas the right conformation is unfavorable due to the steric hinderence cause by the gauche positioning of the ethyl and methyl groups*

Keep in mind that elimination is also **stereospecific** as shown in the following figure.

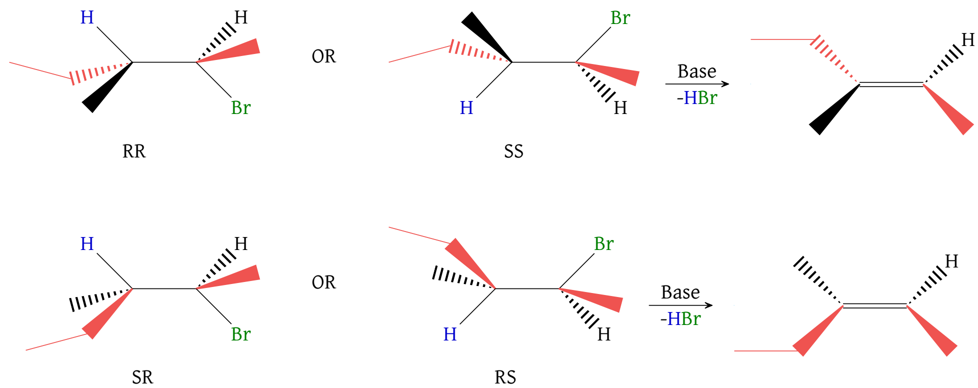


Figure 13: Stereospecificity of HX elimination

Core Concept 1.5: Alkenes from ROH By Dehydration

Alkenes can also be synthesized by exposing alcohols to acids and subsequently removing the resulting H_2O byproduct through dehydration.

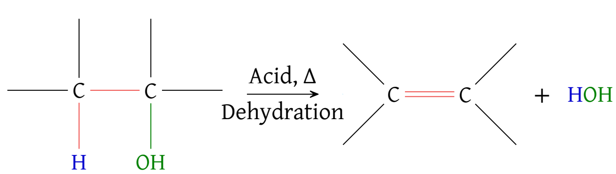


Figure 14: Formation of an alkene via alcohol dehydration

The mechanism behind the reaction—using sulfuric acid and ethanol as an example—is as follows.

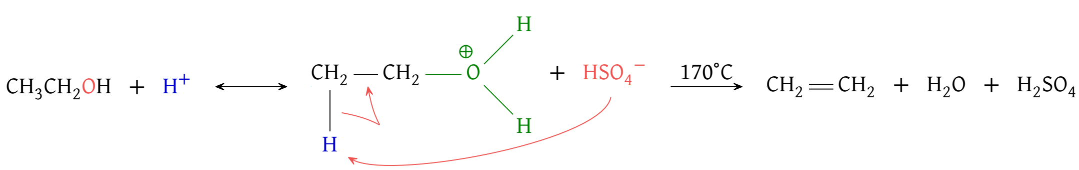


Figure 15: Alcohol dehydration example using H_2SO_4 and ethanol; HSO_4^- unusually acts as a base in this scenario

Due to the effects of steric hindrance and hyperconjugation, the use of E1 versus E2 and the resulting stereoisomer product ratio differ across $R_{primary}$, $R_{secondary}$, and $R_{tertiary}$.

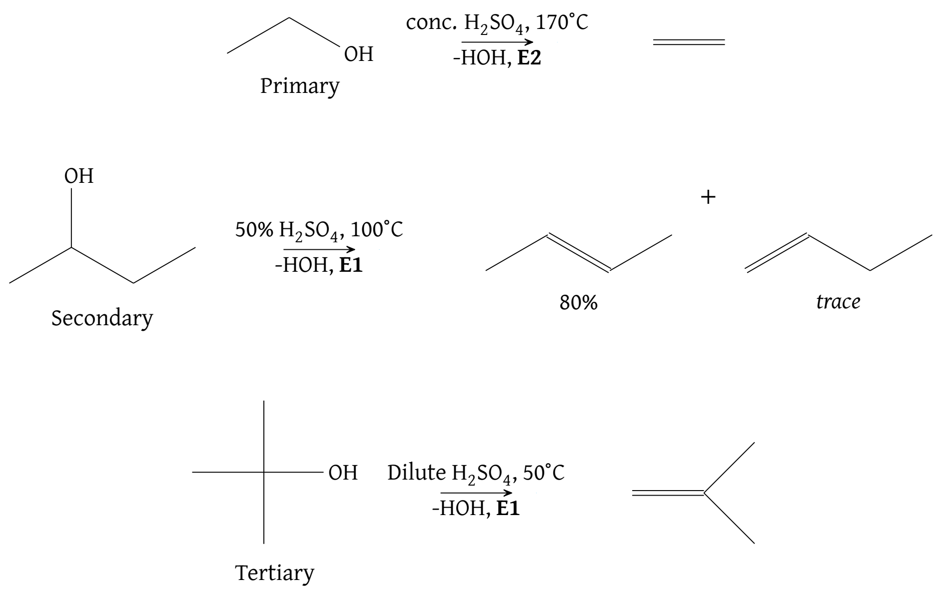


Figure 16: Alkene product ratios from alcohol dehydration across $R_{primary}$, $R_{secondary}$, and $R_{tertiary}$

A major limitation of this approach is the formation of a **carbocation transition state**, which opens the door to **unintended rearrangement pathways** such as **hydride** (*Core Concept 3.1*) or **alkyl shifts** (*Core Concept 3.3*) where possible. As a result, this alkene synthesis pathway has limited utility in practice.

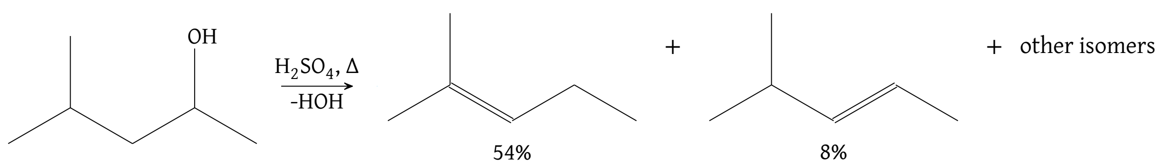


Figure 17: Product mixtures from alcohol dehydration due to hydride shifts

1.2 Alkene Addition Reactions

Core Concept 1.6: Alkene Addition Reactions

Alkenes are the fundamental building blocks for organic synthesis and the preparation of polymer materials. Their defining feature is the **unsaturated π bond**, which reactions through **addition reactions**, wherein the π bond is broken and a diatomic or compound molecule is added across the relevant carbon atoms of the former alkene.

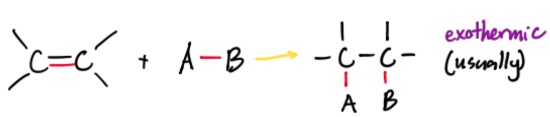


Figure 18

Generally speaking, alkene addition reactions are $\Delta H^\circ < 0$, however since the nature of the reaction is a combination reaction wherein two reactant molecules form a single product molecule, $\Delta S^\circ < 0$, making it entropically unfavorable. In most reactions, the enthalpic favorability overrides the magnitude of entropic unfavorability. However, in some reactions or at high temperatures, the ΔG° turns positive, making these addition reactions thermodynamically unfavorable.

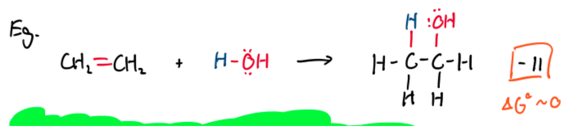


Figure 19

1.2.1 Catalytic Hydrogenation

Core Concept 1.7: Catalytic Hydrogenation

Catalytic hydrogenation refers to the addition of H_2 to an alkene, effectively removing the alkene double bond and converting the target molecule into an alkane if the double bond in question was the only one. This process alone would require enormous amounts of pressure and extreme temperatures to occur, however **catalysts**—such as Pd/C or PtO_2—enable a mechanism with lower E_a that chemists frequently use.

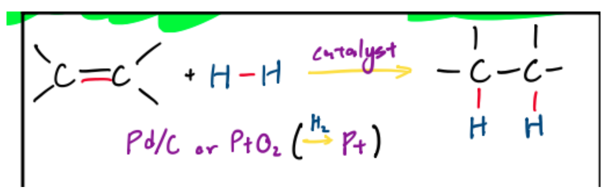


Figure 20

It's also important to note that the hydrogens are **stereospecifically** added to the **same side (syn)** of the target double bond. As a result, the outputed product mixutre will be **racemic**.

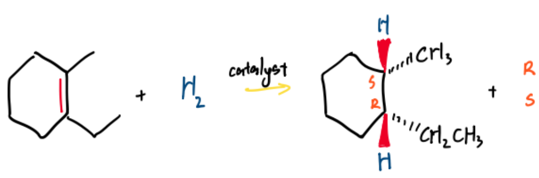


Figure 21

1.2.2 Electrophilic Additions

Core Concept 1.8: Eletrophilic Additions

Because the π bond is **e^- rich**, it readily **reacts with a polar reagent $A^{\delta+}$–$B^{\delta-}$**, where the partially positive end $A^{\delta+}$ is drawn to the e^- rich double bond.

In the scenario where $A^{\delta+}$ is a H^+, the reaction that ensues is simply the reverse of an E1.

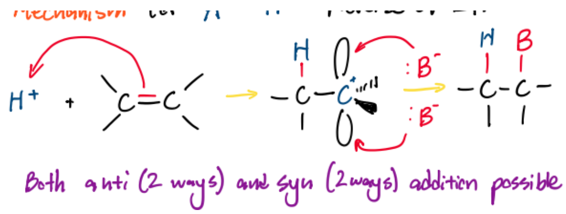


Figure 22: indicate anti and syn stereoisomers in caption

Core Concept 1.9: Hydrohalogenation

Hydrohalogenation is a type of electrophilic addition wherein a **HX is added** to the alkene double bond where X is a halogen.

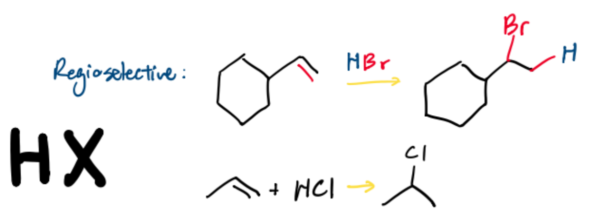


Figure 23

Notice that the H^+ adds to the **less substituted** carbon because it results in a more substituted carbocation in the other carbon, favoring tertiary > secondary > primary carbocation formation. This phenomenon is called **Markovnikov's Rule** and the following energy diagram for the different stereospecific pathways depicts the stability of the different transition states nicely.

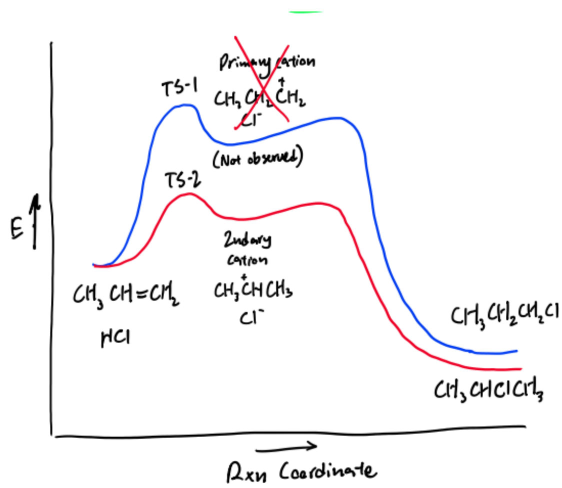


Figure 24

Core Concept 1.10: Markovnikov Hydration

Markovnikov hydration is refers to the conversion of an alkene to an alcohol through the **addition of H_2O** in the presence of an acid catalyst—the reverse of the alkene synthesis reaction covered in Core Concept 1.5.

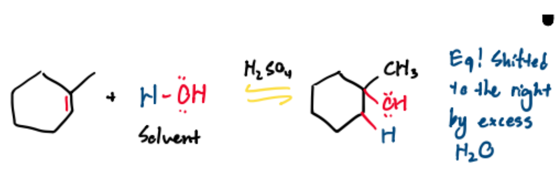


Figure 25

The underlying mechanism to the reaction is shown in the following. Notice how the H^+ is regenerated by the end of the reaction.

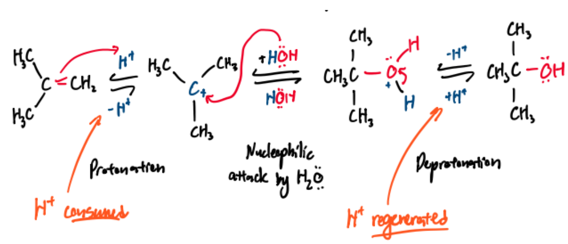


Figure 26

There are, however, bound to be a mixture of alkenes, transient intermediaries, and alcohols because Markovnikov hydration is at dynamic equilibrium with the pathway previously covered in Core Concept 1.5. Because the temporary presence of a carbocation introduces hydride shifts, catalytic H^+ will turn a single alkene species into a slurry of stereoisomeric alkenes, with ratios determined by relative thermodynamic stabilties primarily based on steric hinderance and hyperconjugation.

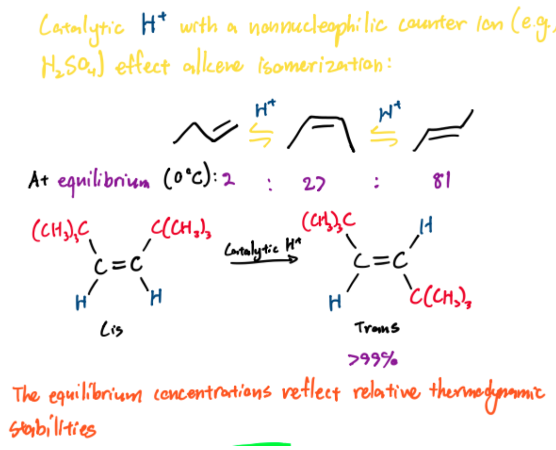


Figure 27

The full mechanism of equilibrium between an alcohol and its corresponding alkene products is shown in the follwing chemical equation using butanol/butene as an example.

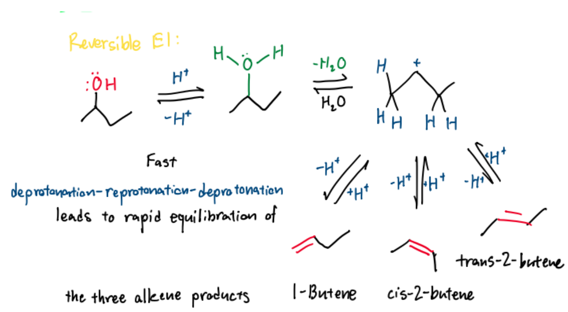


Figure 28: caption includes: leads to the rapid equilibrium of three alkene products

Core Concept 1.11: Halogenation

Halogenation is the addition of a diatomic halogen molecule X_2 to an alkene double bond. In contrast to the addition of HX (*Core Concept 1.9*)—which is inherently polarized due to the electronegativity differences—X_2 is not polarized inherently, but rather **gets polarized during its apporach to the e^- rich alkene double bond**.

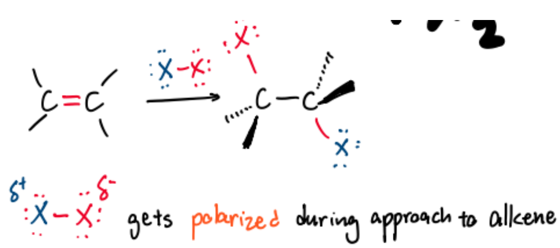


Figure 29

The underlying mechanism is shown in the following figure.

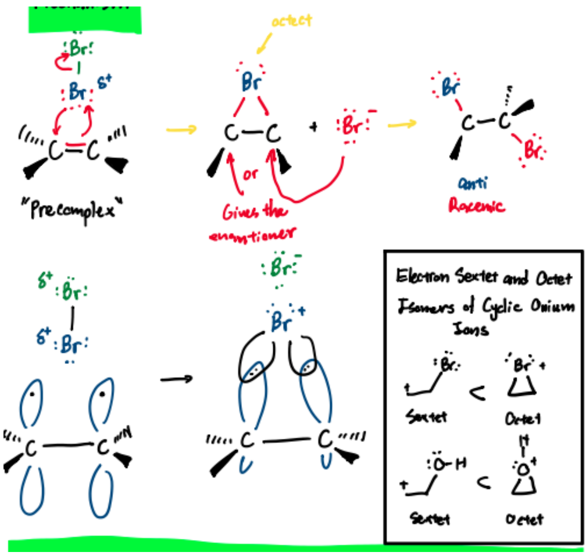


Figure 30

Notice the formation of a halonium ring because the alternative carbocation state will want to distribute the burden of its positive charge. Because the second attack to the bridged halonium exclusively occurs to the backside of the first halogen attack, halogenation is **stereospecifically anti (not syn)**. Likewise, if the second attack has thermodynamically equal possibility of attack either backside carbon—meaning that the inital alkene was perfectly **symmetrical**—the ensuing products will be an **racemic mixture**.

If the initial alkene is **asymmetrical**, the halonium ion intermediate is actually highly distorted, wherein the bond between the halogen and more substituted carbon is **longer**. Likewise, as the halogen pulls more e^- density towards itself the more substituted carbon is able possess a greater δ^+ due to hyperconjugation. As a result, thehalonium ring transition state **already has a high degree of carbocation** behavior and thus mimics more of a S_N1 attack characteristic although the backside attack is S_N2 by definition because of its exclusively anti result. Consequently, the backside nucleophilic attack occurs at the more substituted carbon, an interaction wherein the impact of charge is greater than steric hinderance on regioselectivity.

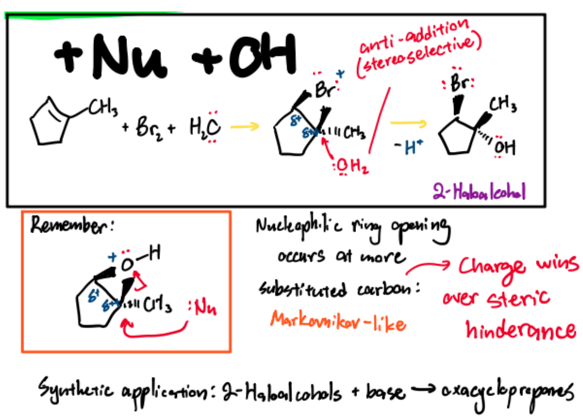


Figure 31

On a different note, the intermediary halonium ring ions can also be intercepted by other nucleophiles such as HOH (*FIG REFERENCE ABOVE*) or ROH (*FIG REFERENCE BELOW*).

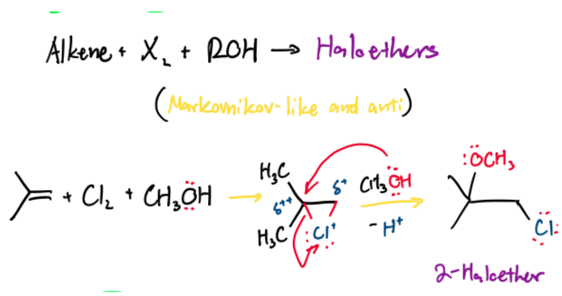


Figure 32

Core Concept 1.12: Oxymercuration–Demercuration

Oxymercuration–demercuration is effectively the same as the **Markovnikov hydration** (*Core Concept 1.10*) process wherein a H_2O is added to remove an alkene double bond. However, oxymercuration–demercuration **avoids carbocation** formation, which prevents the unintended rearranged product alcohols and produces a purer product mix.

The process utlizes mercury (II) acetate and sodium borohydride in a two step chemical process to achieve its H_2O addition.

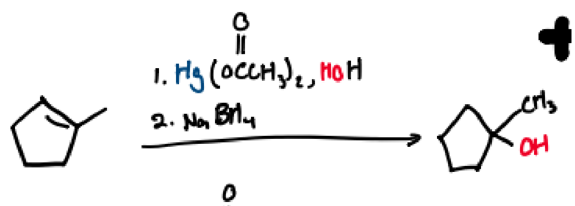


Figure 33

<figure of simple H2O equation>

(1) **Oxymercuration**

Similar to the **alkene halogenation** (*Core Concept 1.11*) process, the large mercury atom forms a bridged, unevenly distributed mercurinium ion rather than a full carbocation, forcing the subsequent H_2O attack to strictly occur at the more substituted carbon.

(2) **Demercuration**

Afterwards, the $NaBH_4$ acts as a reducing agent that cleaves the C–Hg bondand replaces it with a simple hydrogen atom.

Likewise, the usage of ROH allows the addition of a –OR group just as how a –OH was added by H_2O.

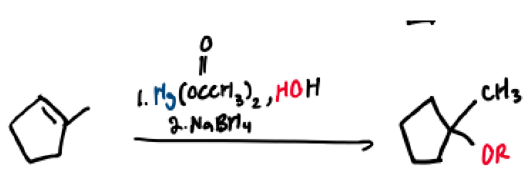


Figure 34

When compared to Markovnikov hydration (*Core Concept 1.10*), oxymercuration–demercuration is the preferred method used by chemists in order to convert an alkene to an alcohol due to the purity of the resulting product mix.

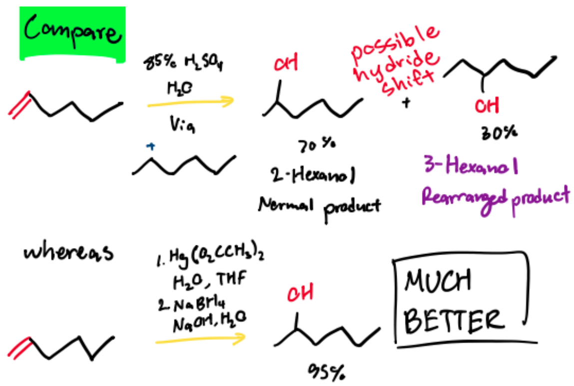


Figure 35

1.2.3 Hydroboration-Oxidations

Core Concept 1.13: Hydroboration-Oxidations

Hydroboration-oxidation is a method chemists use to **add H_2O** to a double bond of an alkene at the **less substituted carbon of the double bond** (*remember that the Markovnikov hydration and oxymercuration-demercuration H_2O additions occur at the more substituted carbon*). This process utilizes BH_3 to regioselectively bond to 3 alkenes and subsequently oxidate and cleave the resulting BR_3 to produce 3 ROH.

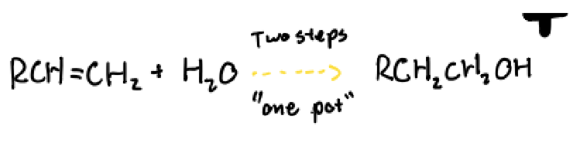


Figure 36

(1) **Hydroboration**

When mixed with BH_3, three alkenes bond to BH_3 with the **less substituted carbon paired to the boron**, occuring due to two major factors.

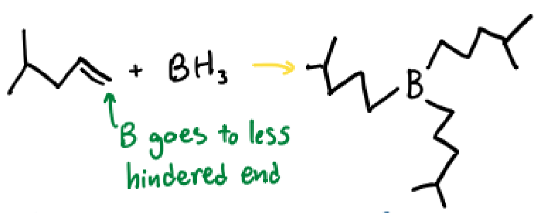


Figure 37

- **Transition State Stabilization**

  When the e^- rich π bond of the alkene reaches out to attack the empty unhybridized p oribital of the boron, the two carbon atoms of the double bond do not share teh burden equally. The carbon that is **not** forming the C–B bond begins to lose its e^- density, creating a partial positive charge δ^+ during its transition state. Consequently, if the less substituted carbon form the C–B bond, the more substituted carbon can better stabilize the this charge through hyperconjugation, allowing for a more stable transition state relative to the alternative C–B bond formation.

<need to make figure>

- **Bond Polarization**

  Another subtle factor that plays a role is the inherent electronegativity difference between boron (2.0) and hydrogen (2.1), creating a polarized bond where boron is slightly positive (δ^+) and hydrogen is slightly negative (δ^-). As a result, the slightly positive boron more favorably pairs with the e^- rich attack from the less substituted alkene carbon and the slightly negative hydrogen more favorably pairs with the subsequently partial positive non-boron bonding alkene carbon.

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Another important characteristic of hydrobromination is the **syn** stereospecificity of the resulting alkylborane. The B and H that are attacked by an alkene π orbital are physcially attached to one another and added to the double bond simultaneously, forcing them to attach to the same face of the alkene.

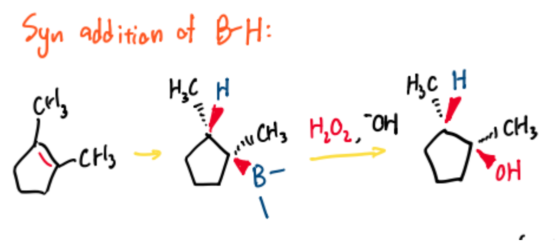

Figure 38

(2) **Oxidation**

The resulting alkylborane R_3B is then oxidized by a solution of H_2O_2 and ^-OH which yields trialkyl borate ($(RO)_3B$) through the following alkyl shift mechanism.

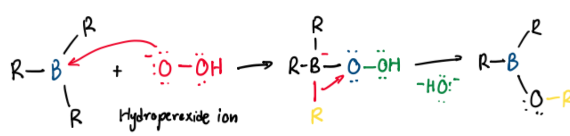

Figure 39

Finally, H_2O hydrolyzes the $(RO)_3B$ with the help of a base to form the intended 3 ROH molecules with a borate ion as a byproduct.

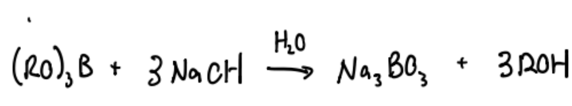

Figure 40

Often this entire process is called **anti-Markovnikov hydration** because it adds a H_2O molecule to the alkene but does so in a manner where the $-OH$ group ends up on the less substituted π bond carbon rather than the more substituted one.

1.2.4 Electrophilic Carbene Additions

Core Concept 1.14: Carbene Synthesis

Carbenes (:CR_2) are sp^2 hybridized **6 e^-** species with a lone pair and empty p orbital—the parent of which is methylene (:CH_2). They can be thought of as being derived by **deprotonating carbocations**.

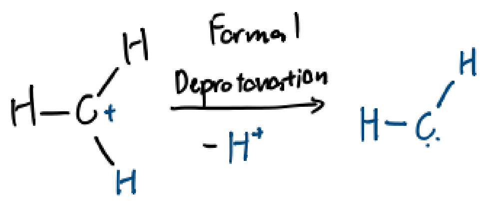

Figure 41

Because they are e^- deficient, carbenes are extermely unstable, short lived intermediates that act as powerful **electrophiles**. The following are the three primary pathways chemists use to generate these intermediates in the laboratory.

(a) **Methylene from diazomethane (CH_2N_2)**

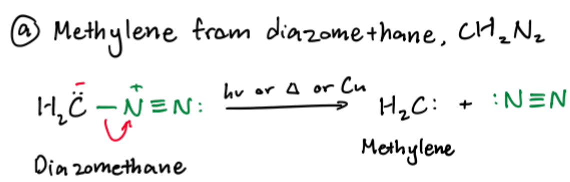

Figure 42

Mind that this reaction is practically irreversible because the resulting N_2 is incredibly stable, driving the reaction forward to cleanly leave behind highly reactive methylene.

(b) **Dichlorocarbene from chloroform**

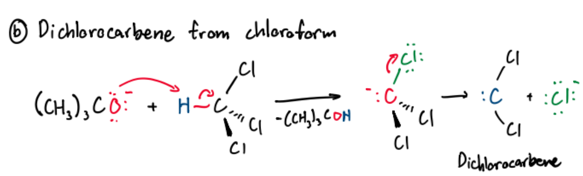

Figure 43

The above reaction utilizes α-elimination (*distinct from the standard β-eliminations E1 or E2 normally used to make alkenes*) through the strong base $(CH_3)_3CO^-$ and the departure of a highly electronegative chlorine atom.

(c) **Simmsons-Smith reagent**

As mentioned previously, free methylene is so wildly reactive that it can be incredibly dangerous to work with. The Simmsons-Smith reaction provides a much safer, controlled workaround.

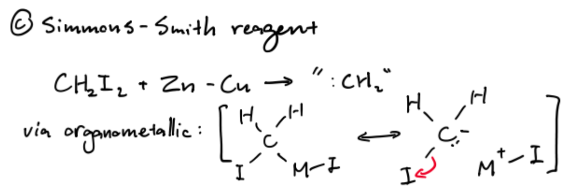

Figure 44

The treatment of diiodomethane with a zinc-copper couple forms an organometallic species that behaves like a hybrid between a stable molecule and a methylene like state.

Core Concept 1.15: Electrophilic Carbene Additions

Being the naturally reactive species that they are, carbenes can be added to alkenes in a **syn stereospecific manner** through the following general mechanism, forming carbon ring as a product.

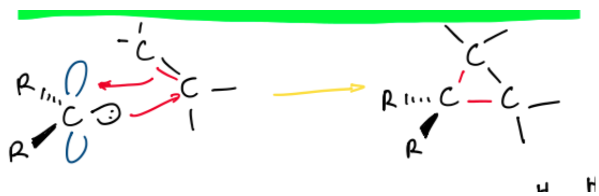

Figure 45

In practice, since carbene species are transient, this carbene addition must occur alongside the carbene synthesis processes (*Core Concept 1.14*).

(a) **Methylene from diazomethane (CH_2N_2)**

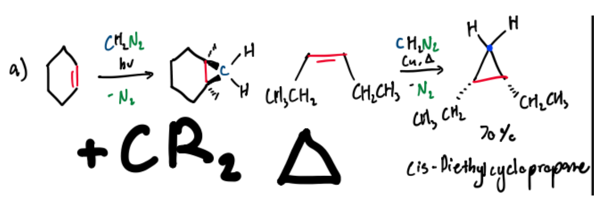

Figure 46

(b) **Dichlorocarbene from chloroform**

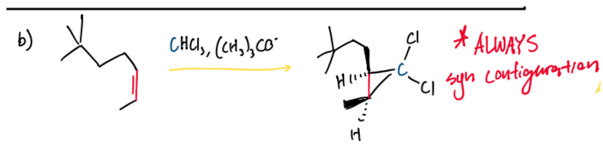

Figure 47

(c) **Simmsons-Smith reagent**

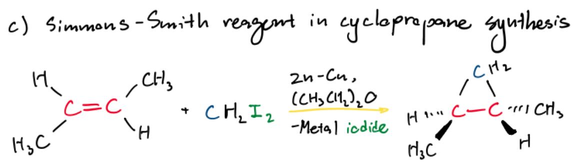

Figure 48

1.2.5 Electrophile Oxidation

Core Concept 1.16: Peroxycarboxylic Acids

Peroxycarboxylic acids are a species of acids with the following general structure.

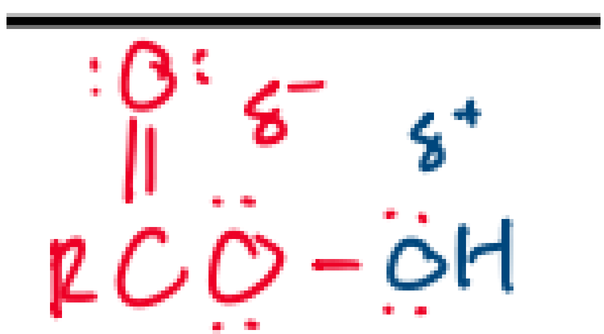

Figure 49: add the commonly used exmaples as well to the figure so it is 4 molecules side by side

When alkenes are treated with these compounds, the reaction results in a **syn addition of an oxygen** that forms a cyclic ether product as outlined below.

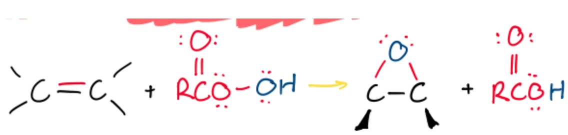

Figure 50: outline with red and blue coloring to represent charges and dashes and wedges to represent the stereospecificity

The underlying **concerted** mechanism looks like the following.

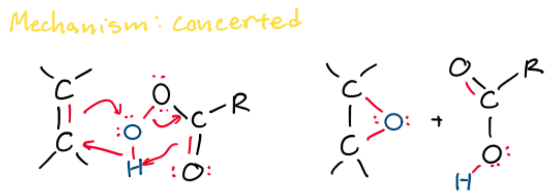

Figure 51

It's also important to note that the **rates of oxacyclopropanation increases with alkyl substitution** because increased hyperconjugation allows to a greater degree of e$^-$ donation from the electrophilic alkene π orbital.

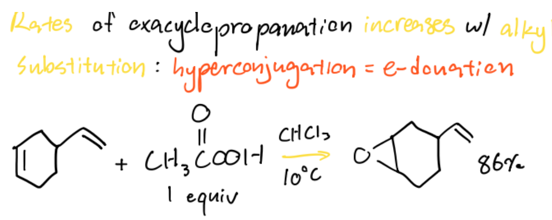

Figure 52

In practice, peroxycarboxylic acid addition usually serves as a prepatory step for subsequent modifications. Two common applcations are shown below.

(a) **Double Nucleophile Syn-Addition**

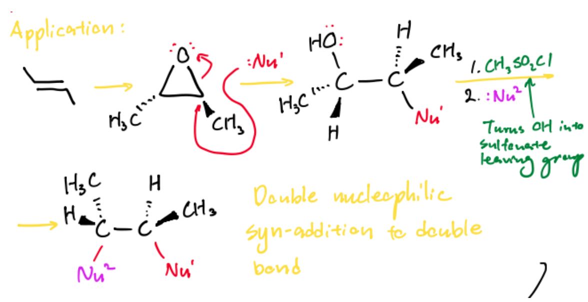

Figure 53

(b) **Anti Stereospecific Dihydroxylation**

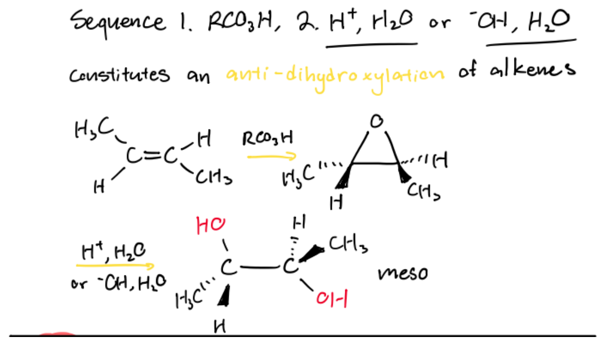

Figure 54

Core Concept 1.17: OsO$_4$ Syn-Dihydroxylation

The treatment of alkenes with OsO$_4$ and a subsequent reduction through an agent like H$_2$S results in syn-dihydroxylation along an alkene double bond.

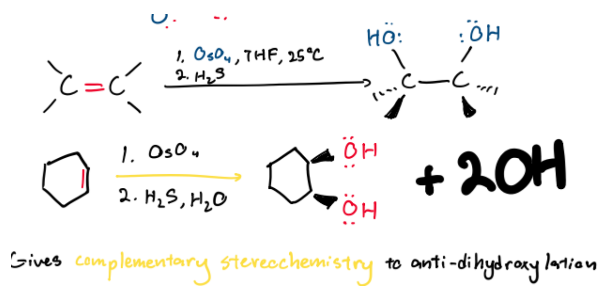

Figure 55

The underlying mechanism proceeds as follows, creating osmate ester intermediary product before the H$_2$S step reduction.

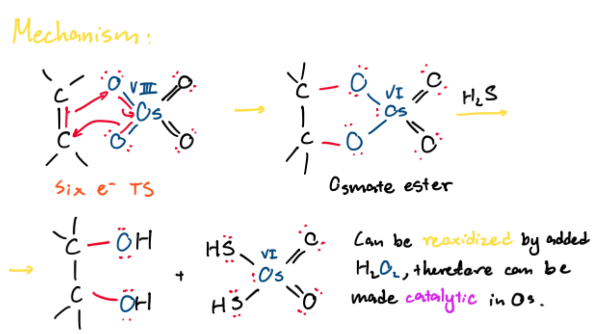

Figure 56

The consequently used Os compound can be reoxidized by adding H$_2$O$_2$, allowing the Os compound to be **catalytic** (*not consumed by the reaction until H$_2$O$_2$ depletes*). This is helpful because Os is expensive.

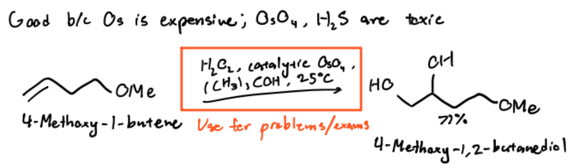

Figure 57

Core Concept 1.18: Ozonolysis

Ozonolysis utlizes ozone (O_3)—a highly unstable and rapidly decaying species—and a reducing agent to perform a complete oxidative cleavage of an alkene double bond.

Figure 58

Because ozone is such a transient molecular specie, it isn't available enough naturally chemists to work with it. To prevent this shortage, chemists put O_2 through an **arc discharge**—continuous, highly intense electrical breakdown of a gas—to generate fresh stream of 2–4% O_3 gas safely in a lab.

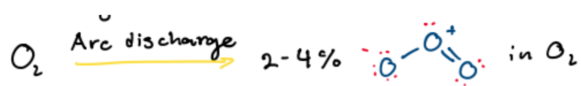

Figure 59

The following are three of the most common reducing agents used by chemists.

(a) **Zinc and Acid (Zn / CH_3COOH)**

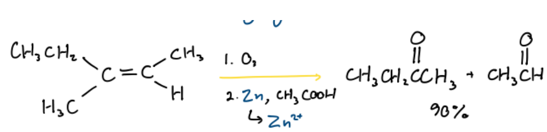

Figure 60

(b) **Catalytic Hydrogenation (H_2 / Pt)**

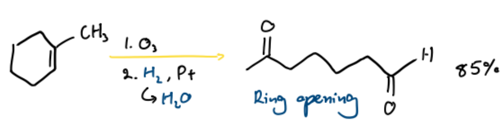

Figure 61

(c) **Dimethyl Sulfide (DMS; $(CH_3)_2S$)**

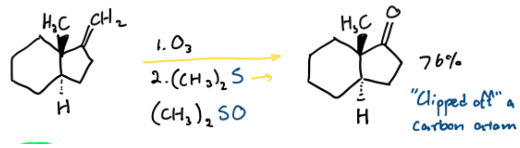

Figure 62

1.2.6 Radical Hydrobromination

Core Concept 1.19: Radical Hydrobromination

Radical hydrobromination refers to the **addition of an HBr** molecule to an alkene double bond with the assistance of RO – OR in an **anti-Markovnikov manner**, meaning that the Br is attached to the **less substituted carbon**. Recall that regular **hydrohalogenation** (*Core Concept 1.9*) follows Markovnikov regiochemistry where the halogen is attached to the more substituted carbon.

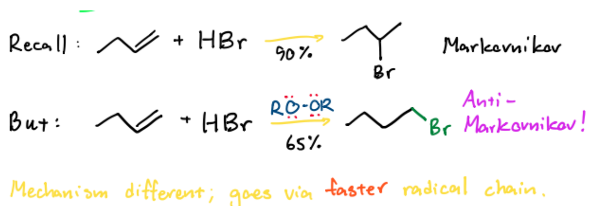

Figure 63

The underlying mechanism is shown in the following figure, started by peroxide radical initiators (*the O–O bond is weak at around* 39 *kcal* mol^{-1})

(a) **Initiation Steps**

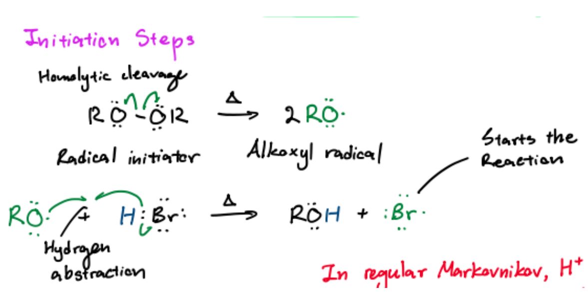

Figure 64

(b) **Propagation Steps**

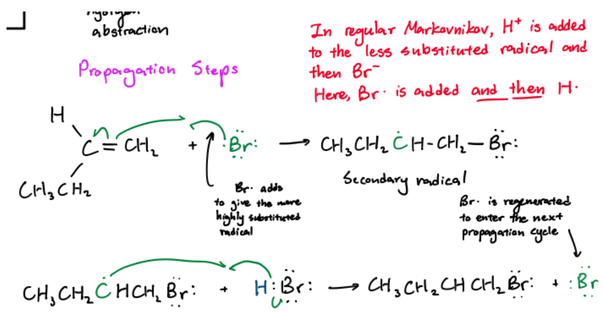

Figure 65: seperate propagation I and II in the first

> Notice that the Br· is added to the less substituted carbon, resulting in the resulting carbon radical to be better hyperconjugated than the alternative binding arrangement due to being more substituted.

This radical hydrobromination does not work for HCl or HI because one of the propagation steps are **endothermic** due the unique electronegative properties of Cl and I. Hence the radical mechanism is too slow and the ionic mechanism in the Markovnikov manner (*attaching to more substituted carbon*) is favored. Br, on the other hand, sits at the thermodynamic sweet spot between Cl and I. RSH anti-markovnikov addition works in the same manner as Br, sitting in that same sweet spot.

2 Alkynes

Core Concept 2.1: Alkyne Orbital Structure

An alkyne triple bond is composed of two perpendicular π bonds and a conjoined sp hybrid σ bond.

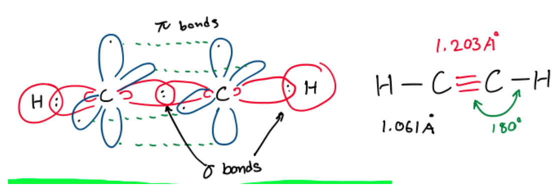

Figure 66

Core Concept 2.2: Triple Bond Strength

Ethyne releases a massive $-311\,\text{kcal mol}^{-1}$ of heat when combusted. Compared to ethane (CH_3CH_3)—which releases −373 kcal/mol—ethyne releases a proportionally greater amount of energy on a per carbon basis because you don't have to spend extra energy to break the 4 additional C–H bonds in ethane.

$$HC{\equiv}CH + 2.5\,O_2 \longrightarrow 2\,CO_2 + H_2O \qquad \Delta H^\circ = -311\,\text{kcal mol}^{-1}$$

$$CH_3CH_3 + 3.5\,O_2 \longrightarrow 2\,CO_2 + 3\,H_2O \qquad \Delta H^\circ = -373\,\text{kcal mol}^{-1}$$

Due to the higher level of strain alkyne π bonds exist under, the full hydrogenation of the two alkyne π bonds releases more energy than that of two alkene π bonds (*which would be roughly* -60 *kcal mol*$^{-1}$)

$$(\textit{terminal})\ CH_3CH_2C{\equiv}CH + 2\,H_2 \xrightarrow{\text{catalyst}} CH_3CH_2CH_2CH_3 \qquad \Delta H^\circ = -69.9\,\text{kcal mol}^{-1}$$

$$(internal)\ CH_3C\equiv CCH_3 + 2\,H_2 \xrightarrow{\text{catalyst}} CH_3CH_2CH_2CH_3 \qquad \Delta H^\circ = -65.1\ \text{kcal mol}^{-1}$$

Notice the differences in net energy release due to the greater stability of the internal triple bond over the terminal version because of hyperconjugation.

Core Concept 2.3: Relative Acidity of Alkynes

In general hydrocarbons (alkanes and alkenes), the C–H bond is considered completely non-acidic. However, terminal alkyne C–H bonds defy this rule when treated with a sufficiently strong base, wherein the terminal H^+ can be plucked off.

$$RC\equiv C-H + B^- \longrightarrow R-C\equiv C^- + HB \qquad pK_a = 25$$

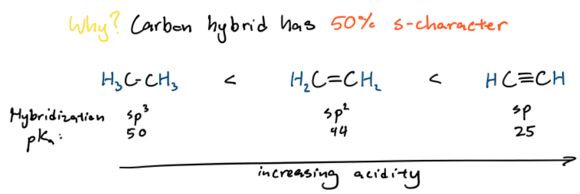

Figure 67

The above acidic trend is explained by the greater s orbital characteristics of the corresponding orbital hybridization. Since an s-orbital is spherical and sits much closer to the positively charged carbon nucleus than a directional, elongated *p*-orbital,the lone pair of electrons left behind after deprotonation is held very close to the positive nucleus. This electrostatic proximity stabilizes the negative charge immensely.making the corresponding alkyne conjugate base comparatively more stable.

Using this alkyne acidity, chemists are able to execute the following example applications

(a) **Organolithium Reagents**

Figure 68

(b) **Sodium Amide ($NaNH_2$)**

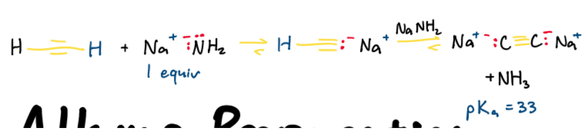

Figure 69

2.1 Alkyne Synthesis

Core Concept 2.4: Elimination E2 From Dihaloalkanes

Performing E2 elimination twice on dihaloalkenes with adjacent halogen substitutents allows for alkyne formation.

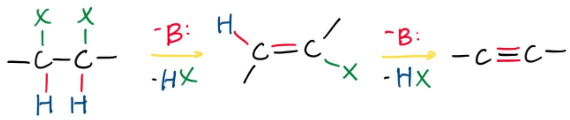

Figure 70

Here is a specific example utlizing $NaNH_2$ as the base, which has to go through an extra aqueous work up step due the acidity of terminal alkynes (*Core Concept 2.3*) in the presence of sodium amide $NaNH_2$.

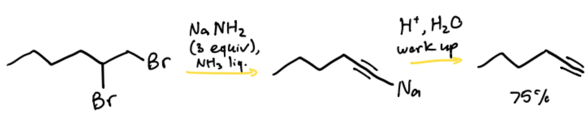

Figure 71

Core Concept 2.5: Alkylation of Alkynyl Metals

Terminal alkynes can be converted into powerful nucleophiles via deprotonation with strong bases, enabling C–C bond formation through substitution pathways that follow strict S_N2 rules.

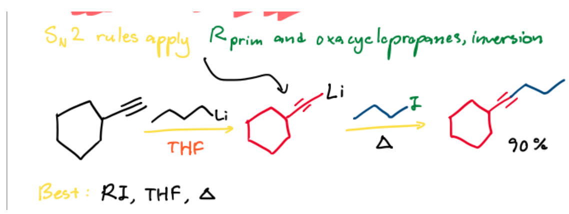

Figure 72

The synthetic utility of alkynyl metals relies on R_{prim} halides or oxacyclopropanes (*epoxides*) as electrophilic substrates to minimize competing E2 elimination side-reactions. When utilizing unsymmetrical epoxides, the acetylide nucleophile regioselectively attacks the less sterically hindered carbon atom.

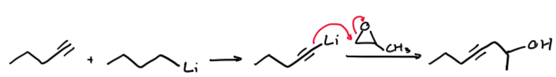

Figure 73

The following are various examples using this mechanism.

(a) **Carbonyl Addition with Grignard**

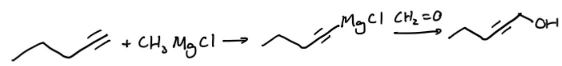

Figure 74

(b) **Opoxide Opening with Lithium Acetylide**

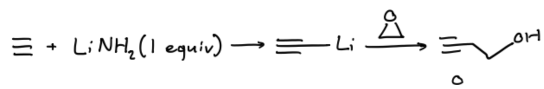

Figure 75

(c) **Double Grignard Addition of Acetaldehyde**

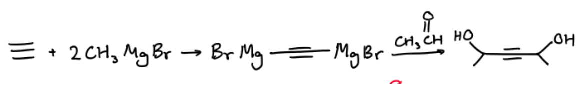

Figure 76

2.2 Alkyne Reactions

2.2.1 Reductions

Core Concept 2.6: Complete Hydrogenation

The complete hydrogenation of an alkyne triple bond (*addition of* $2H_2$) progresses similarly alkene hydrogenation (*Core Concept 1.7*), utilizing a catalyst like Pt to help lower the E_a of the reaction. The resulting product is an alkane.

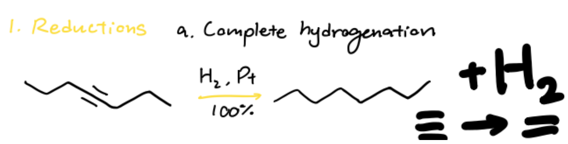

Figure 77

Core Concept 2.7: Partial Hydrogenation

Partial hydrogenation refers to the addition of only one H_2 wherein only one of the two π bonds are removed, resulting in either cis or trans product from a syn and anti addition, respectively.

- **Lindler's Catalyst (Syn Addition)**

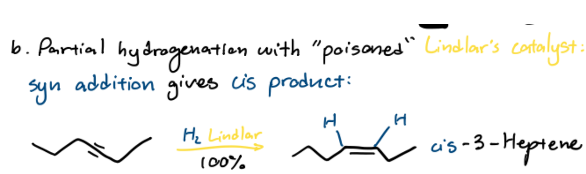

Figure 78

A syn addition and subsequent cis product requires the usage of a "*poisoned*" **Lindlar's catalyst**.

- **Sodium Reduction (Anti Addition)**

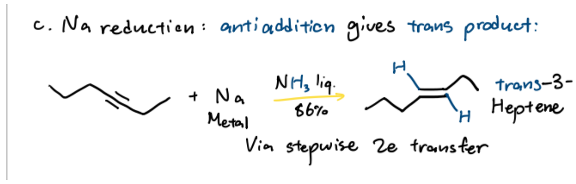

Figure 79

The underlying mechanism for the above reduction through sodium is shown below.

(1) **e⁻ Transfer**

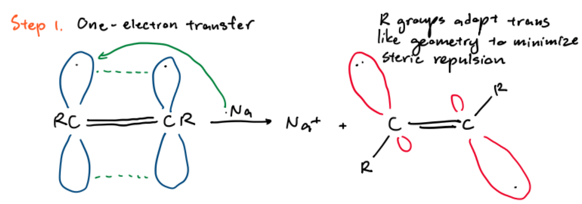

Figure 80

(2) **First Protonation**

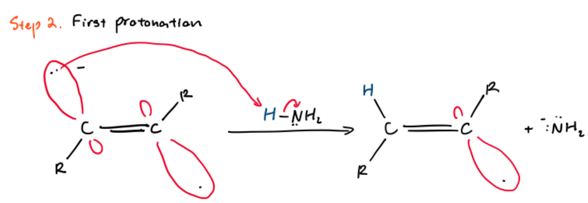

Figure 81

(3) **Second e⁻ Transfer**

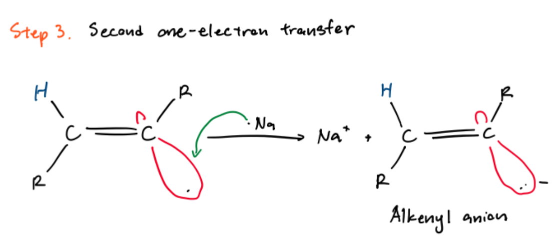

Figure 82

(4) **Second Protonation**

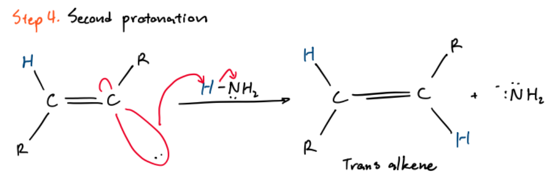

Figure 83

2.2.2 Electrophilic Additions (Markovnikov)

Core Concept 2.8: HX Anti Addition

The **addition of HX** to an alkyne mix converts the triple bond to a **geminal** (*same side*) dihaloalkane.

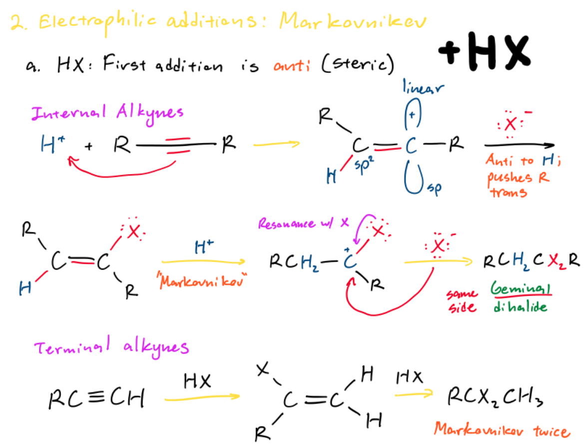

Figure 84

It's important to notice that the first HX addition is always exceuted in the anti configuration because of steric hinderance.

Additionally, although FIGURE REFERENCE categorizes this HX addition process into internal and terminal alkyne categories, it's much simpler to understand that the ensuing transition states of the reaction favor carbocation formation on the more substituted carbon (*or Markovnikov's rule*). Because of this tendency, the second HX addition will bind the second H^+ to the same first H^+ receiving carbon because the alternative carbon is better substittued by the presence of a halogen substitutent from the first HX addition.

Here are some real reactions, showcasing alkyene HX anti additions.

(a) **Symmetrical Internal Alkyne**

Figure 85

(b) **Asymmetrical Internal Alkyne**

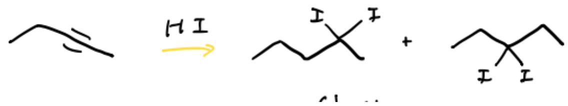

Figure 86

Here because the degree of hyperconjugation between the bonded R groups do not have a large difference, mixtures ensure although the PRODUCT NAME most likely is more favored.

(c) **Terminal Alkyne**

Figure 87

(d) **Synthetic Application (Vicinal to Geminal)**

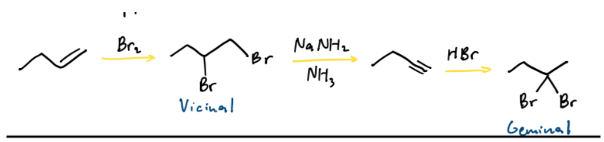

Figure 88

The first step of the chemical reaction of above is alkene halogenation, covered in Core Concept 1.11.

Core Concept 2.9: X_2 Anti Addition

Alkyne halogenation **adds X_2** to an alkyne triple bond, resulting in the addition of four halogens (*two to each triple bond carbon*) via a transient ring state. The mechanism for these additions an anti arrangement and is effectively alkene halogenation (*Core Concept 1.11*) occuring twice in a row.

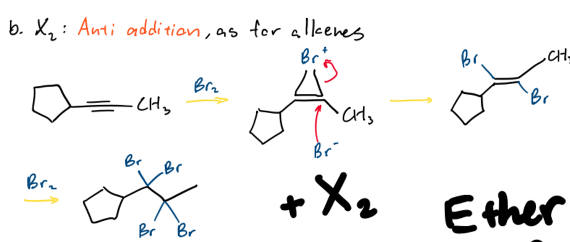

Figure 89

Core Concept 2.10: Mercury Catalyzed Markovnikov Hydration

The treatment of an alkyne with catalytic $HgSO_4$ (*previously seen in Core Concept 1.12*) and H_2O leads to an unstable enol intermediate in a Markovnikov manner. The subsuquent **taurtomerization** of this enol with catalytic H^+ or ^-OH leads to the formation of a ketone, regardless of whether the alkyne triple bond is terminal or internal (*because the ^-OH group is attached to the more substituted carbon.*).

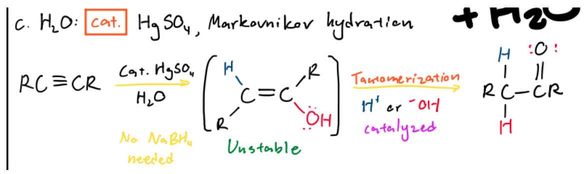

Figure 90

The underlying mechanism of **taurtomerization**—both H^+ and ^-OH pathways—is shown below.

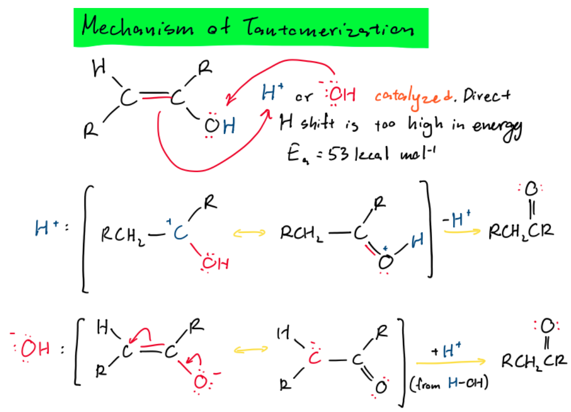

Figure 91

Here are also some practical use cases in the laboratory.

(a) **Asymmetrical Internal Alkyne**

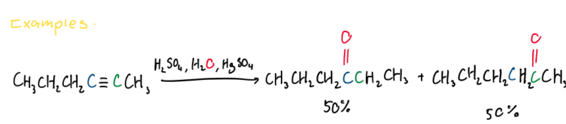

Figure 92

(b) **Terminal Alkyne**

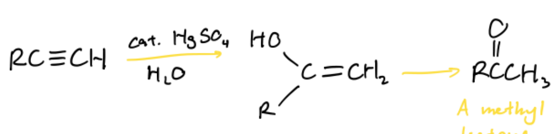

Figure 93

Core Concept 2.11: Radical Hydrobromination

Radical hydrobromination in alkynes is the **single addition of HBr** through the formation of radical species with the assistance of ROOR. The reaction follows the same mechanism covered in the alkene version of this electrophilic addition, covered in Core Concept 1.19, however leads to the formation of product mixes due to the possible rotation along the resultingly formed double bond.

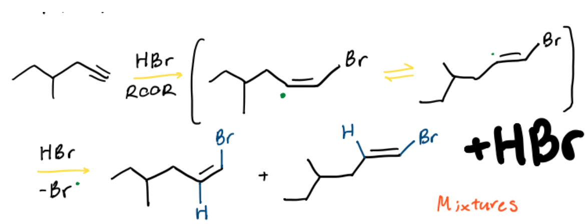

Figure 94

Unlike the additions we've seen in HX anti addition (*Core Concept 2.8*) and X_2 anti addition (*Core Concept 2.9*), this reaction does not nessecarily undergo a second iteration to convert the resulting alkene further into an alkane form. This is because the mechanism requires the presence of an e^- dense π bond, which the resulting bromine-attached alkene does not satisfy because bromine is highly electronegative, pulling e^- density away from the double bond (*a clean alkene does not experience this*). As a result, because the starting alkyne is significantly more reactive than the bromoalkene product, the radicals will preferentially consume all the remaining alkyne before even attempting to touch the bromoalkene intermediate.

If one exposes alkynes to an excess of HBr and peroxides, however, a second radical addtion can occur, which follows the same Markovnikov stability rules to form a **geminal dihaloalkane**, the same product of HX anti addition (*Core Concept 2.8*), making the extra peroxide use unnessecary if the orignal intention was to form a geminal dihaloalkane.

Core Concept 2.12: Hydroboration-Oxidation

Alkyne **hydroboration-oxidation** utilizes a R_2BH (*R is a bulky group such as cyclohexyl*) to ensure the hydroboration of only one π bond. *Recall that alkene hydroboration-oxidation (Core Concept 1.13) utilized simply BH_3 because there is only one π bond to begin with.* The mechanism follows the same process as the alkene version of this reaction (*Core Concept 1.13*) and effectively accomplishes an anti-Markovnikov hydration.

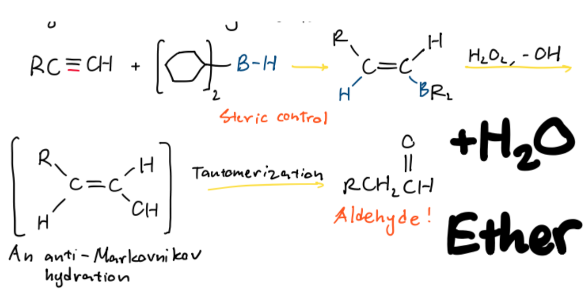

Figure 95: exclude the taut step here

The resulting alkenol can subsequently be put through previously covered processes like taurtomerization to be convereted into an aldehyde.

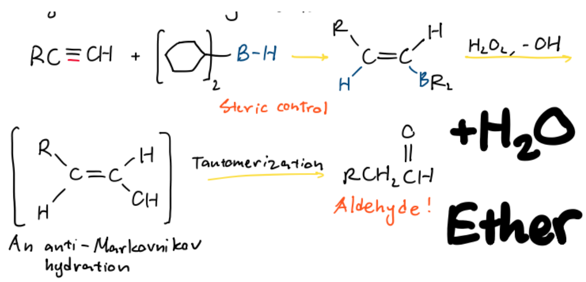

Figure 96: only the taut step here

The following is a direct comparison with **mercury catalyzed Markovnikov hydration** (*Core Concept 2.10*), which has a similar but distinct outcome resulting in a ketone rather than an aldehyde.

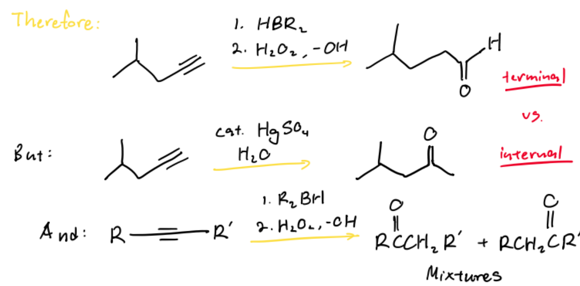

Figure 97

In an internal alkyne scenario, mixtures ensue when the degrees of substitution in the triple bond carbons don't have a large difference (*although the slightly less substituted side will be favored in the context of final carbonyl positioning*).